

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	7.5
NN	7.625
Sweep	7.75
SaveSeq	5
SaveParall	2.625
MJ	1
CMT	2.375
Kilby	5.75
SaveMatch	7
FJ	8.375

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 9 degrees of freedom): 54.709091.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
9	FJ	4.871765	0.000001	0.005556	0.005683	0.033488
8	Sweep	4.458903	0.000008	0.00625	0.011334	0.033488
7	NN	4.376331	0.000012	0.007143	0.016952	0.033488
6	Random	4.293759	0.000018	0.008333	0.022539	0.033488
5	SaveMatch	3.96347	0.000074	0.01	0.028094	0.033488
4	Kilby	3.137747	0.001703	0.0125	0.033617	0.033488
3	SaveSeq	2.642313	0.008234	0.016667	0.039109	0.033488
2	SaveParall	1.07344	0.283074	0.025	0.04457	0.033488
1	CMT	0.908295	0.363722	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.025 .

Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.04457 .

Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.033488 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	FJ	0.00001	0.00001
2	Sweep	0.00008	0.00066
3	NN	0.00012	0.00084
4	Random	0.00018	0.00105
5	SaveMatch	0.00074	0.00369
6	Kilby	0.001703	0.00681
7	SaveSeq	0.008234	0.024703
8	SaveParall	0.283074	0.566148
9	CMT	0.363722	0.566148

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	FJ	0.00001	0.00001	0.000002
2	Sweep	0.00008	0.000037	0.000013
3	NN	0.00012	0.000037	0.000019
4	Random	0.00018	0.00004	0.000028
5	SaveMatch	0.00074	0.00133	0.000116
6	Kilby	0.001703	0.002553	0.002669
7	SaveSeq	0.008234	0.010574	0.012776
8	SaveParall	0.283074	0.312285	0.307906
9	CMT	0.363722	0.363722	0.363722

Table 4: Adjusted p -values (FRIEDMAN) (II)